

A03 Bridging Neuroscience and Artificial Intelligence

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1. Neural Network Comparison Analysis

Biological vs. Artificial Neurons

Biological neurons are the basic building blocks of the human brain. They were the inspiration for the artificial neurons employed in today's AI systems. A biological neuron has dendrites that get signals from outside the neuron, a cell body (soma) that processes those signals, an axon that sends the output, and synapses that control how strong the connections between neurons are. Artificial neurons also take in information, use weighted connections, do math on it, and then send it out.

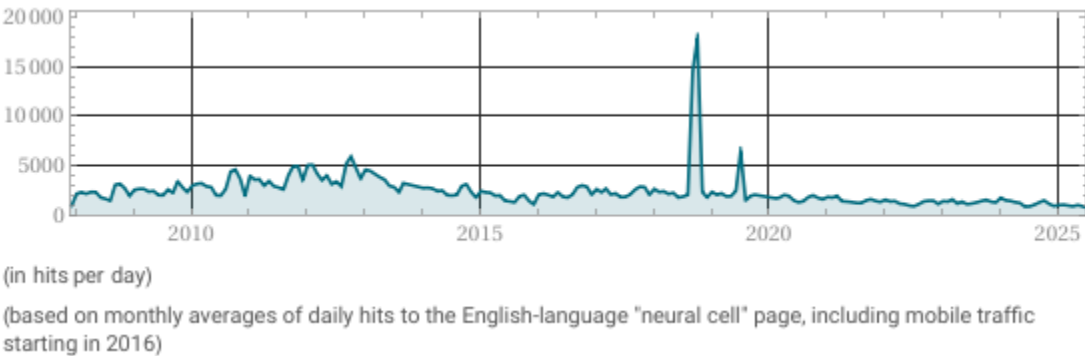
This biological inspiration led to the creation of artificial neural networks by adding ideas like weighted inputs, activation functions, and layered processing. But there are important disparities in how they are put into action. Biological neurons talk to each other using electrochemical signals, change all the time through synaptic plasticity, and use energy very efficiently. Artificial neurons, on the other hand, use numbers to do math and backpropagation, and therefore need a lot of computing power. Artificial neural networks are great at speed, scalability, and accuracy for certain tasks, but they don't have the adaptability, robustness, or general intelligence of biological systems.

Comparative Analysis Table

Biological Neural Networks	Artificial Neural Networks	AI Architecture Examples
Neurons with dendrites and axons	Artificial neurons with weighted inputs and outputs	Convolutional Neural Networks (CNNs)
Synaptic plasticity	Trainable weights via backpropagation	Deep Neural Networks
Hierarchical brain organization	Layered network structures	CNNs, Transformers
Parallel biological processing	GPU-based parallel computation	Deep learning models
Energy efficient (~20 watts)	High energy consumption	Large-scale AI systems

Figure 1. Neuron anatomy showing dendrites, soma, axon, and synapses.

Source: Wolfram Alpha (“neuron anatomy”)



2. Brain Architecture and AI Design

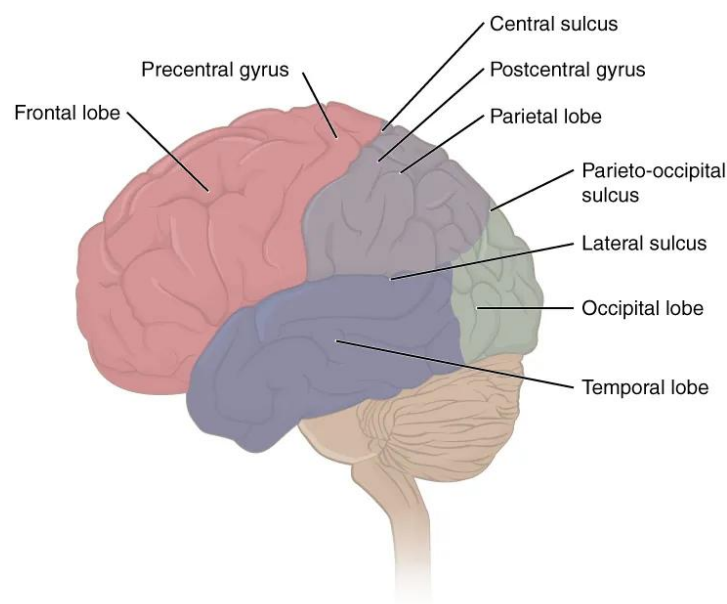
The brain is arranged in a hierarchy, with different parts of the brain doing different things. Sensory areas deal with raw data, intermediate areas find patterns, and higher-level areas do think and make decisions. This hierarchical structure is very similar to how modern AI works, with the first levels finding simple traits and the later layers finding more abstract representations.

Convolutional Neural Networks (CNNs) act like the visual cortex by processing visual information in steps, finding edges, textures, and forms before putting together whole

object representations. Long Short-Term Memory (LSTM) networks are like hippocampus because they store information over time and help people learn by using their memories. Attention processes in transformer models are similar to biological attention systems in that they give priority to relevant information and ignore signals that aren't.

The amygdala is very important for processing emotions, figuring out risks, and making decisions. Learning about this biological function could help AI systems by making them better at figuring out risks, learning on their own, and recognizing patterns in context. AI that is inspired by emotions, could help people make better decisions in situations where they don't know what to do and make it easier for people and AI to talk to each other.

Figure 2. Major regions of the human brain, including the frontal, parietal, temporal, and occipital lobes, illustrating functional specialization and hierarchical organization (Source: OpenStax Anatomy & Physiology).



3. Integration Analysis and Future Implications

Learning about biological brain networks can help us create better artificial intelligence systems. Biological systems have extraordinary efficiency, flexibility, and generalization characteristics that contemporary AI systems continue to find challenging to replicate. Researchers can create new learning algorithms that don't need huge, labeled datasets by looking at how real neurons learn through local plasticity and self-organization.

Neuromorphic computing and spiking neural networks are examples of biomimetic techniques that try to copy the brain's efficiency while still being able to do math.

There are several contrasts between biological and artificial systems. Biological minds learn and adapt to new situations all the time while using very little energy. Artificial systems are faster and more accurate for little jobs, but they need a lot of retraining and use a lot of energy. When it comes to generalization, biological intelligence is still better since people can learn from only a few examples and apply what they know to other areas.

Future AI development may benefit from a more thorough incorporation of biological ideas. Neuroscience-inspired solutions could help with problems including catastrophic forgetting, poor generalization, and high energy needs. Hybrid systems that mix biological knowledge with artificial computation could make intelligent systems that are stronger, more flexible, and more efficient. This would move AI research and use forward.

References

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